

Databases for data analytics

<https://github.com/evidencebp/databases-course/>

Window Functions

Window Functions Goal

- Usually we look at a single raw record
- Grouping allows us to look on groups of records
- Window functions allows us to look at a single row and “look through the window” on the context, other rows in the query dataset

Window definition

- Current record (implicit yet very important)
- Name (in order to reuse it)
- Partition (the relevant grouping)
- Order (sorting the window)
- Frame (context filtering)

Window is a (rather) good metaphor

- Current record - floor
- Name (e.g., left, center, right)
- Partition (**bad metaphor**
 , maybe sides)
- Order (going up or down)
- Frame (which neighboring
 floors can we see)

Given a floor,
specific window functions compute
a value on the floor.



How to use windows functions

Decide what are the anchor rows that you would like to enhance

- They might be computed with views, group bys, etc

Decide on the window context

- Partition - in which column the anchor row is similar to the windows' rows
- Order - what is the timeline, if any, on which the window's rows should be ordered
- Frame - rows filtering by the relative position to the anchor on the timeline
- Name - to improve comprehension and reuse

As in group by, windows may contain many rows.

- Use aggregation and window functions to select a single value
- You can apply many functions to the same window

Syntax - Rolling average

SELECT product, year, month, volume,

avg(volume) OVER w AS rolling_3m_avg **# Avg aggregation as the window function**

FROM sales

WINDOW w AS (**# Name - the common name "w"**

PARTITION BY product **# Partition - only other rows of the same product**

ORDER BY year, month **# Order chronologically, defining before and after**

ROWS BETWEEN 2 PRECEDING AND CURRENT ROW **# Frame - 3 last months**

)

ORDER BY product, year, month; **# Order the anchor rows - First by product, later by the same timeline**

Common uses

Usually applied to aggregated data in BI systems

- Looking at a month before/after
- Rolling metric (e.g., last 3 month average)
- Specific value (instead of max, the 10th value)
- Percentiles
- Relative position

Resources

- [Window functions](#)
- Sales [examples](#)
- See IMDB [examples](#)

Windows functions implementation has sharp edges and database differences (e.g., count distinct, relations to group by). It is out of the course scope yet remember to use carefully.